



First & Last Touch Attribution

Technical Architecture and Operations Guide

A clear reference for consent-aware capture, attribution state, propagation, and controlled maintenance.

This guide describes how First & Last Touch Attribution works with standard Odoo Website, CRM, and Sales architecture. It is intended for Odoo administrators, implementation partners, developers, support engineers, and technical owners who need to understand browser capture, signed state, data models, propagation rules, security, reliability, and operational checks.

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SECTION Technical Purpose and Scope

The module adds a first-party acquisition layer to the standard Odoo Website-to-CRM-to-Sales flow. It does not replace Website routing, the standard UTM subsystem, CRM creation, customer creation, or Sales. It captures one First Touch snapshot and one Latest Qualified Acquisition snapshot, then propagates them only through explicit supported lifecycle points.

The central technical principles are:

- browser capture respects Odoo optional-cookie consent;
- the canonical browser state is signed, website-specific, and host-specific;
- First Touch is immutable after initialization;
- Latest Qualified Acquisition changes only for qualified touches;
- each touch is atomic and never borrows missing values from another visit;
- Website form stamping is server-side and needs no hidden attribution fields;
- new customer and opportunity-to-quotation propagation are explicit lifecycle hooks;
- existing customers and duplicated business records are protected from false attribution;
- native Odoo campaign/source/medium behavior remains independent and unchanged;
- public and non-sales users do not receive access to the attribution fields.

TECHNICAL NAME website_first_last_touch

SECTION Component Map

Component	Technical responsibility
controllers/attribution.py	Public CSRF-protected POST endpoint for explicit capture after optional-cookie consent.
models/attribution_mixin.py	Field definitions, snapshot building, sanitization, qualification, state merge, signing, cookie I/O, form capture, and record snapshot helpers.
models/ir_http.py	Post-dispatch hook for eligible frontend HTML GET responses.
models/crm_lead.py	Website form stamping, lead merge rules, new-customer propagation, and quotation context propagation.
models/res_partner.py	Adds the common attribution fields to customers/contacts.
models/sale_order.py	Adds attribution fields, quote/Sales creation reconciliation, and copy-context protection.
static/src/interactions/attribution_consent.js	Consent-event capture and timing compatibility.
views/*.xml	CRM, customer, and Sales tabs, optional list columns, search fields, filters, and group-by controls.
tests/test_attribution.py	TransactionCase coverage for algorithms and business-model propagation.
tests/test_http.py	HttpCase coverage for consent, real HTTP capture, Website forms, and native UTM.
static/tests/attribution_consent.test.js	Unit tests for capture request construction and consent-triggered capture.

Manifest dependencies are website_crm and sale_crm. No external Python package or external attribution SaaS is required by the module.

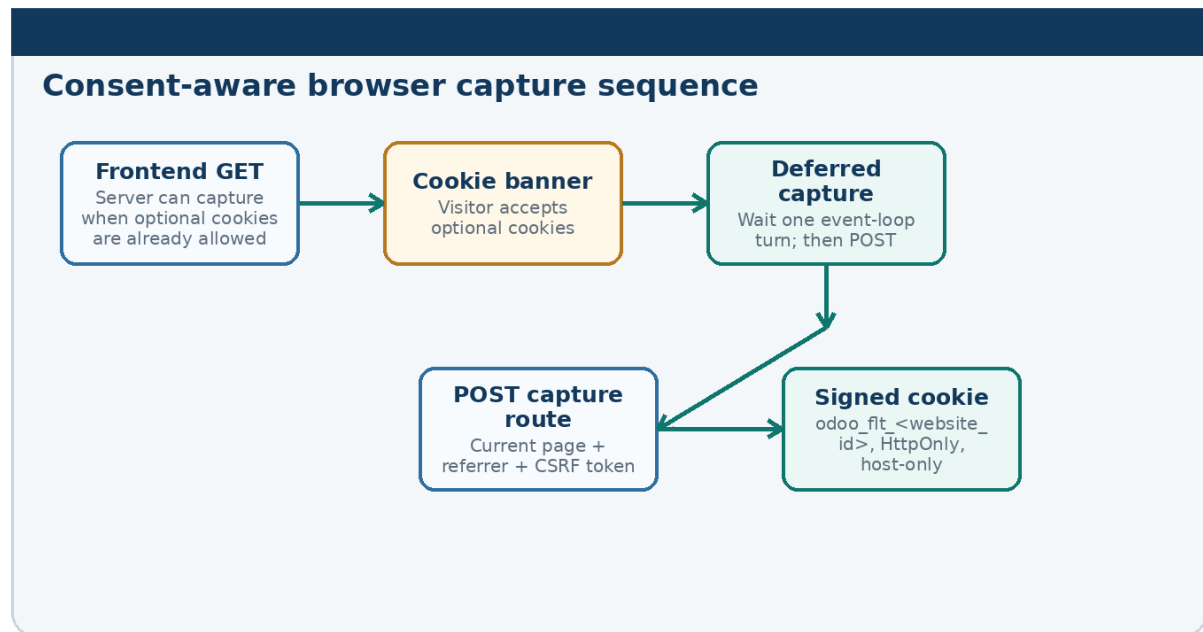
SECTION Capture Entry Points

Attribution can be captured through three coordinated server paths and one browser event path. They share the same engine, so validation and merge semantics remain consistent.

Entry point	When it runs	Purpose
ir.http._post_dispatch	Eligible frontend HTML GET response with HTTP 200	Normal server-side page capture when optional cookies are already allowed.

Entry point	When it runs	Purpose
/website/first-last-touch/capture	Explicit POST after consent	Captures the current page after optional cookies are accepted on a page that was already rendered.
crm.lead.website_form_input_filter	Standard Website CRM form submission	Captures current/referrer context and stamps the resulting CRM values in the same request.
optionalCookiesAccepted / modal fallback	Browser event after the Website cookie decision	Starts the explicit POST at the correct time and prevents duplicate sends.

SECTION Consent Event and JavaScript Lifecycle



Consent-aware explicit capture sequence

The frontend script imports the standard Odoo HTTP post helper and keeps one module-level pendingCapture promise to deduplicate the consent flow.

Event path

1. The script listens once for document event optionalCookiesAccepted.
2. captureAttributionAfterConsent checks whether a capture is already pending.
3. The request is deferred with setTimeout(..., 0). This lets the standard Website cookie-bar code continue its hide sequence and persist website_cookies_bar before the capture POST is sent.
4. The POST sends window.location.pathname + window.location.search, document.referrer, and the current CSRF token.
5. A hidden.bs.modal listener on the Website cookie bar is kept as a compatibility fallback. It removes itself and calls the same deduplicated capture function.
6. Network failure is caught so the optional attribution request does not break the Website interaction.

WHY THE DELAY EXISTS The server checks the incoming website_cookies_bar cookie before it allows an optional attribution cookie. Sending the POST before the consent cookie is persisted would return HTTP 200 but still produce no attribution cookie. The deferred request closes that timing gap.

SECTION Capture Controller

The public route is /website/first-last-touch/capture. It uses type=http, auth=public, website=True, POST only, CSRF protection, and readonly=True.

The controller creates a JSON response with {"ok": true}, truncates page input to 4,096 characters and referrer input to 2,048 characters, then delegates to `_attribution_capture_explicit()`. The response status is not itself proof that a cookie was written; the engine can intentionally refuse capture when optional cookies are not allowed or the input is invalid.

Security properties

- POST only;
- CSRF token required;
- public route does not expose business records;
- all browser page/referrer input is treated as untrusted and revalidated by the server.

SECTION Frontend Response Capture

`ir.http._post_dispatch` loads the final response and calls `_attribution_process_response()`. Normal page capture is eligible only when all conditions are true:

- `request.is_frontend` is true;
- HTTP method is GET;
- response status is 200;
- response MIME type is text/html;
- the request path does not start with `/web/` or the explicit capture endpoint.

Before any capture, `_attribution_optional_cookies_allowed()` delegates to `ir.http._is_allowed_cookie("optional")`. If optional cookies are not allowed, an existing attribution cookie is deleted through a required-cookie response so withdrawal can be enforced.

SECTION Attribution State and Snapshot Schema

Cookie state

Key	Meaning
version	State schema version. Current value is 1.
website_id	Website ID used to isolate state.
host	Normalized current host used to reject cross-host reuse.
first	Complete First Touch snapshot. Required for a valid stored state.
latest	Complete Latest Qualified Acquisition snapshot or false.

Snapshot

Key	Type	Meaning
at	Datetime string	Server observation time.
kind	String	direct, utm, ad_click, or referral.
landing_path	String	Normalized page path, maximum 512 characters in a valid snapshot.
referrer	String	External origin or empty string.
utm	Object	Allowlisted source, medium, campaign, term, content values.
click_ids	Object	Allowlisted gclid, gbraid, wbraid, fbclid, msclkid values.

When a snapshot is applied to a business record, it is expanded with `version`, `website_id`, and `host` so the stored business snapshot remains self-describing.

SECTION Touch Classification

`_build_attribution_snapshot()` first validates the current page against the current host, extracts allowlisted query values, reduces the referrer to an external origin when appropriate, and then assigns one kind.

Priority	Condition	kind
1	At least one supported click ID exists	ad_click
2	No click ID, but at least one UTM value exists	utm
3	No marketing parameter, but valid external referrer exists	referral
4	No marketing parameter and no external referrer	direct

One important exception exists: if there are no marketing values and the referrer is same-site, the result is `None` rather than `Direct`. This treats ordinary internal navigation as no new touch.

SECTION First and Latest Merge Semantics

`_attribution_merge_state()` validates the existing state before it is used. Invalid state is replaced with a clean state bound to the current website and host.

First Touch

If no first snapshot exists, the candidate becomes First Touch. This write is permanent for the browser state. Direct is accepted as a valid First Touch.

Latest Qualified Acquisition

The candidate can update Latest only when `_is_qualified_attribution_touch()` is true and its fingerprint differs from the current latest fingerprint.

Fingerprint

The fingerprint contains kind, landing_path, UTM object, and click_ids object. The external referrer is included only when no marketing UTM/click parameter exists. Therefore a marketing touch with identical parameters and landing is considered the same even when an incidental referrer changes.

Business-record merge

`_merge_attribution_snapshots()` selects the earliest valid snapshot when `latest=False` and the latest valid qualified snapshot when `latest=True`. The complete JSON object is copied; fields are not mixed between snapshots.

SECTION Input Sanitization and URL Safety

Value normalization

- Unicode is normalized with NFKC.
- Characters in Unicode control categories are removed.
- Whitespace sequences are collapsed and trimmed.
- `utm_source` and `utm_medium` are capped at 100 UTF-8 bytes.
- `utm_campaign` and `utm_term` are capped at 150 UTF-8 bytes.
- `utm_content` is capped at 200 UTF-8 bytes.
- click IDs are capped at 256 UTF-8 bytes and must match `[A-Za-z0-9._~-]+`.
- query parsing is limited to 50 fields; only supported parameter names are accepted; the first occurrence of each accepted name wins.

Host and path handling

- hosts are normalized to lowercase IDNA form and trailing dots are removed;

- a current page URL with a foreign host is rejected;
- malformed URLs that raise parsing errors are treated as empty and rejected;
- paths are normalized to start with /, repeated slashes are collapsed, and the normal limit is 512 UTF-8 bytes;
- external referrer must use http or https and is reduced to scheme://host.

SECTION Cookie Signing, Retention, and Compaction

The cookie name is `odoo_ftl_<website_id>`. The default cookie domain is `None`, which keeps it host-only. Retention is 31 days.

Write attributes

Attribute	Value / rule
<code>max_age</code>	31 days
<code>expires</code>	Current UTC time + retention
<code>path</code>	/
<code>domain</code>	None (host-only)
<code>secure</code>	True when the current request is HTTPS
<code>httponly</code>	True
<code>samesite</code>	Lax
<code>cookie_type</code>	optional

Signing

State is signed with Odoo `hash_sign` under the module scope `website_first_last_touch.cookie` and the same retention period. Decoding uses `verify_hash_signed`. A tampered token, oversized token, wrong website ID, wrong host, invalid snapshot, wrong type, or expired signature is rejected.

Size control

The hard token limit is 3,500 characters. Snapshots are compacted when necessary: UTM values and landing paths receive tighter limits, secondary click IDs can be dropped in aggressive mode, and content/term can be removed if required. If the final signed token still does not fit, the engine deletes the attribution cookie rather than storing an invalid state.

SECURITY NOTE Signing provides integrity, not confidentiality. Anyone who can read the raw cookie value outside `HttpOnly` browser protections should not assume the marketing data is encrypted.

SECTION Website Form Integration

`crm.lead.website_form_input_filter()` calls the standard implementation first. The module then captures the Website form request into `request.future_response` and updates the create values with `_attribution_values_from_state()`.

Same-request recovery

A Website form POST does not necessarily contain the original landing URL.

`_attribution_capture_website_form()` examines the request URL and Referer. If the form POST contains no supported marketing parameter and the Referer is on the same host, the referrer page is used as `page_url` and the referrer itself is cleared. This lets a standard Contact Us form preserve the acquisition page in the same request without hidden fields.

The request environment caches one form state under `website_first_last_touch.form_state` so repeated calls in the same request use the same result.

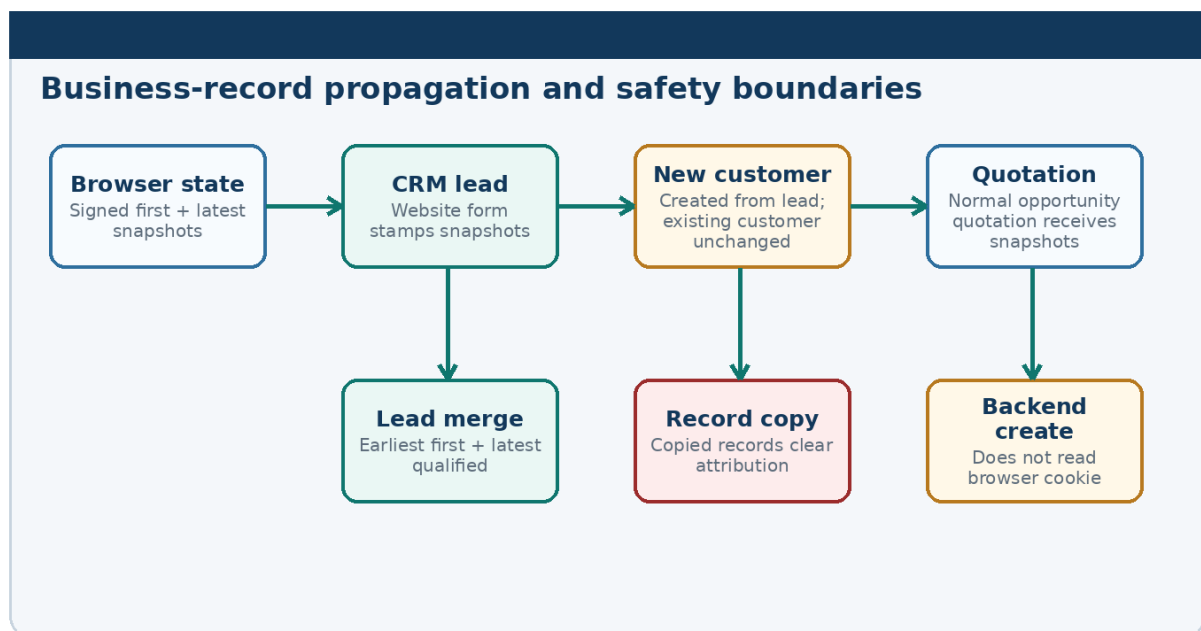
SECTION Business Data Model

The abstract model `website.first.last.touch.mixin` defines two canonical JSON fields and stored summary fields. `crm.lead`, `res.partner`, and `sale.order` inherit the mixin.

Field	Type	Technical role
<code>flt_first_touch_data</code>	Json	Canonical First Touch snapshot. readonly, copy=False, Sales group.
<code>flt_latest_touch_data</code>	Json	Canonical Latest Qualified Acquisition snapshot. readonly, copy=False, Sales group.
<code>flt_first_touch_at</code> / <code>flt_latest_touch_at</code>	Datetime	Stored computed summary timestamps.
<code>flt_first_source</code> / <code>flt_latest_source</code>	Char	Stored source summaries; both indexed when non-empty.
<code>flt_first_medium</code> / <code>flt_latest_medium</code>	Char	Stored medium summaries.
<code>flt_first_campaign</code> / <code>flt_latest_campaign</code>	Char	Stored campaign summaries; latest campaign is indexed when non-empty.
<code>flt_first_term</code> / <code>flt_latest_term</code>	Char	Stored term summaries.
<code>flt_first_content</code> / <code>flt_latest_content</code>	Char	Stored content summaries.
<code>flt_first_landing_path</code> / <code>flt_latest_landing_path</code>	Char	Stored landing-path summaries.
<code>flt_first_referrer</code> / <code>flt_latest_referrer</code>	Char	Stored external-origin summaries.
<code>flt_first_click_id_type</code> / <code>flt_latest_click_id_type</code>	Char	Readable primary click-ID type.
<code>flt_first_click_id</code> / <code>flt_latest_click_id</code>	Char	Readable primary click-ID value; latest click ID is indexed when non-empty.
<code>flt_has_ad_click_id</code>	Boolean	True when first or latest snapshot contains a readable supported click ID.

Summary computation first validates the JSON snapshot. Invalid data produces empty summary fields instead of trusting malformed structures.

SECTION CRM Integration



Propagation and false-attribution safety boundaries

Website form stamping

The Website CRM form path updates create values with `flt_first_touch_data` and, when present, `flt_latest_touch_data`. A qualified First Touch is also exported as Latest Qualified Acquisition when no later latest snapshot exists. A Direct-only First Touch exports no latest field.

Lead merge

`crm.lead._merge_get_fields_specific()` registers custom aggregation functions. First Touch uses the earliest valid complete snapshot. Latest uses the latest valid qualified complete snapshot.

UI

The CRM form contains an Attribution tab for Sales users. Lead and opportunity list views add optional summary columns. CRM search views add first/latest search fields, Attributed and Has Ad Click ID filters, and group-by First Source, Latest Source, and Latest Campaign.

SECTION Customer Propagation

`crm.lead._create_customer()` records existing partner IDs before delegating to the standard customer-creation logic. After the standard method returns, attribution is written only when the returned partner is newly created in this flow, was not an existing partner/parent candidate, and does not already have First Touch data.

This rule protects shared or existing customers from being rewritten by a new lead. `res.partner` itself only inherits the common fields; it has no automatic browser-cookie read.

The customer Acquisition tab is Sales-group restricted and invisible when First Touch data is empty. The latest column is labeled Latest at Customer Creation to describe its freeze semantics.

SECTION Sales Propagation

Opportunity quotation context

`crm.lead._prepare_opportunity_quotation_context()` adds deep-copied `default_ftl_first_touch_data` and `default_ftl_latest_touch_data` values only when they are not already present in the context. This is the normal opportunity-to-quotation propagation path.

sale.order.create() reconciliation

The create override intentionally does not pull attribution only because `opportunity_id` exists. Reconciliation runs only when an opportunity exists and at least one attribution field is already present in the incoming values. When it runs, a missing First Touch can be filled from the opportunity and Latest is selected from the explicit and opportunity snapshots by latest qualified server timestamp.

Copy safety

`sale.order.copy()` removes `default_ftl_first_touch_data` and `default_ftl_latest_touch_data` from context before calling the standard copy. Together with `copy=False` on the fields, this prevents stale quotation defaults from reintroducing attribution into a duplicate record.

SECTION Lead Merge and Copy Safety

Operation	Rule
Merge CRM leads	Earliest complete First Touch + latest complete qualified Latest Acquisition.
Copy CRM lead	Both JSON attribution fields clear because <code>copy=False</code> .
Create new customer from lead	Copy lead snapshots only to the newly created customer.
Assign existing customer	Do not overwrite existing customer attribution.
Copy customer	Attribution clears because <code>copy=False</code> .
Create quotation from opportunity	Normal opportunity context supplies snapshots.
Copy quotation / Sales Order	Attribution clears; stale default context is stripped.
Plain backend <code>sale.order</code> create	Does not read browser state and does not invent attribution.

SECTION Native Odoo UTM Coexistence

The module never writes `campaign_id`, `source_id`, or `medium_id` directly in its business propagation hooks. Standard Website UTM handling continues independently. Automated HTTP tests confirm that `NativeSource`, `NativeMedium`, and `NativeCampaign` still populate the native relational fields while the module also stores its latest source summary.

This separation is intentional. Native UTM fields remain suitable for standard Odoo reporting. The prefixed `flt_*` fields are the canonical atomic first/latest context supplied by this module.

SECTION Security and Access Control

All common attribution fields use `groups="sales_team.group_sale_salesman"`. This group restriction is applied at field level, so `fields_get` for a public user or an internal non-sales user does not expose the protected attribution fields.

Controller and request security

- public capture route is POST-only and CSRF-protected;
- browser page/referrer inputs are size-limited and validated;
- signed state validates schema, website ID, and host;
- the module does not expose an RPC endpoint to read attribution business records.

Record security

The module does not add permissive record rules for CRM, contacts, or Sales. Standard application permissions continue to control which business records a Sales user can open.

SECTION Search Fields and Indexing

The stored computed summary fields allow normal ORM search and group operations. Four high-value fields use `btree_not_null` indexes:

- `flt_first_source`
- `flt_latest_source`
- `flt_latest_campaign`
- `flt_latest_click_id`

The boolean `flt_has_ad_click_id` supports the Has Ad Click ID filter. The Attributed filter uses `flt_first_touch_at != False`. Search views expose First Source, Latest Source, Latest Campaign, and Latest Click ID as explicit search fields.

SECTION Privacy and Data Minimization

The implementation deliberately avoids building an identity profile. It does not read or store IP address, user agent, browser fingerprint, Website form email, or Website form phone for attribution.

URL storage is minimized: landing query strings are not preserved; external referrer path/query/fragment are discarded; only allowlisted UTM/click parameters are processed. Signed browser state is a short-lived transport and acquisition-memory mechanism, while business-record retention follows the normal retention of the CRM/customer/Sales record.

IMPORTANT The cookie signature prevents undetected modification but is not encryption. Business owners should avoid placing personal data or secrets in campaign values or public paths.

SECTION Performance Considerations

The normal GET path adds capture work only for frontend HTML 200 responses after optional cookies are allowed. The capture engine performs bounded query parsing, bounded string

normalization, one signed-cookie decode, snapshot comparison, and a cookie write only when state changes or invalid state must be replaced.

Repeated identical qualified touches do not rewrite the cookie or refresh the latest timestamp. This reduces unnecessary browser-state churn.

Business-record summary fields are stored computed fields, so list/search/group operations do not parse the JSON snapshot for every row. High-value search fields are indexed.

No external API call is part of page capture, Website form stamping, customer propagation, or Sales propagation.

SECTION Deployment, Upgrade, and Recovery

Deployment sequence

1. Back up the database and add-on files according to the normal release process.
2. Place `website_first_last_touch` in the configured add-ons path.
3. Restart the Odoo service.
4. Update the Apps list and install or upgrade the module through the normal registry process.
5. Review Website cookie-bar settings and the target Website form action.
6. Run the campaign-to-cookie-to-CRM acceptance flow.
7. Run opportunity-to-quotation, customer creation, copy safety, and security checks.
8. Monitor application logs and browser console/network output during the first controlled test cycle.

Recovery

If browser attribution must stop immediately, disable or change optional-cookie consent behavior at the Website policy level or uninstall/disable the custom module according to the normal change process. Existing browser cookies become inert when the module is no longer present and expire according to their remaining lifetime. Business-record fields follow normal module-uninstall database behavior.

Domain changes

Because signed state validates the host, a domain or reverse-proxy host change should be followed by a fresh browser acceptance test. Old host-bound state is not trusted on the new host.

SECTION Operational Troubleshooting

Symptom	Meaning	Technical check
Capture POST returns 200 but no <code>odoo_ft_cookie</code>	Server refused optional-cookie write because consent cookie was not yet present or optional cookies are not allowed.	Check <code>website_cookies_bar</code> on the request; confirm deferred post-consent capture is loaded.
Malformed page produces no snapshot	Expected URL safety behavior.	Inspect supplied page/referrer and confirm a valid same-host page path.
Cookie decodes as invalid	Signature, schema, website ID, host, size, or expiration check failed.	Discard state and start a clean acquisition visit.
Form creates lead without attribution	No allowed browser state and no same-request acquisition context was available.	Check consent, form Referer, campaign URL, and whether the form uses <code>website_crm</code> .
Lead merge result looks partial	Latest snapshot may itself be a partial qualified touch.	Inspect raw first/latest JSON; partial snapshots intentionally stay atomic.
Existing partner not updated	Expected safety rule.	Use lead attribution as the historical acquisition; do not overwrite shared customer data.
Copied quotation unexpectedly has data	Would indicate copy-context regression.	Confirm copy override strips <code>default_ft_*</code> context and rerun regression tests.
Non-sales user can see fields	Security regression.	Check field groups and inherited views; rerun <code>fields_get</code> security checks.

SECTION Extension Boundaries

The base module is intentionally narrow. Separate extensions can build on the canonical snapshots and summary fields without weakening the base semantics.

- offline conversion export to advertising platforms;
- ad spend and ROAS dashboards;
- multi-touch journey ledger;
- channel classification and attribution models;
- eCommerce checkout attribution;
- Events attribution;
- external Website/form connectors;
- cross-device identity;
- custom retention or domain policy through supported method overrides;
- additional allowlisted marketing parameters through `_attribution_parameter_names()` and related sanitization logic.

Extensions should preserve the two core guarantees: First Touch remains stable, and Latest remains one complete qualified snapshot rather than a mixture of values from different visits.

SECTION Technical Acceptance Checklist

- Module installs/upgrades through the normal registry process.
- Frontend asset contains `attribution_consent.js` and unit-test assets load.
- Optional-cookie consent allows capture; missing/declined consent does not.
- Deferred consent capture writes the signed attribution cookie in a real browser.
- Cookie is website-specific, host-specific, signed, HttpOnly, host-only, SameSite=Lax, and retained for the configured period.
- UTM and click-ID allowlists, size limits, control-character stripping, and invalid-character rejection work.
- Landing query is discarded and referrer is reduced to external origin.
- Same-site navigation is not a new touch.
- First Touch is immutable and Direct does not replace Latest.
- Repeated identical qualified touch does not refresh timestamp.
- Standard Website CRM form needs no hidden attribution fields.
- Native Odoo campaign/source/medium remains functional.
- Lead merge keeps earliest first and latest qualified snapshots.
- New customer receives lead attribution; existing customer is not overwritten.
- Opportunity quotation receives attribution.
- Copied lead/customer/quotation clears attribution.
- Plain backend business creation does not read browser cookie.
- Public/non-sales users cannot access attribution fields; Sales user can.
- Search fields, filters, group-by controls, and indexed summaries are available.

SECTION Support Information

Technical name: `website_first_last_touch`

Dependencies: `website_crm`, `sale_crm`

License: LGPL-3

Support contact: full.stack.odoo@gmail.com

For technical investigation, provide the request path, public host, website ID, consent state, whether the attribution cookie exists, expected and actual first/latest summaries, business record model and ID, how the record was created (Website form, customer creation, quotation flow, copy, or backend integration), and the relevant server/browser log section. Remove session cookies, passwords, tokens, and unrelated private customer content.